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Addressing Discovery and Topology Challenges in M-PESTI Systems

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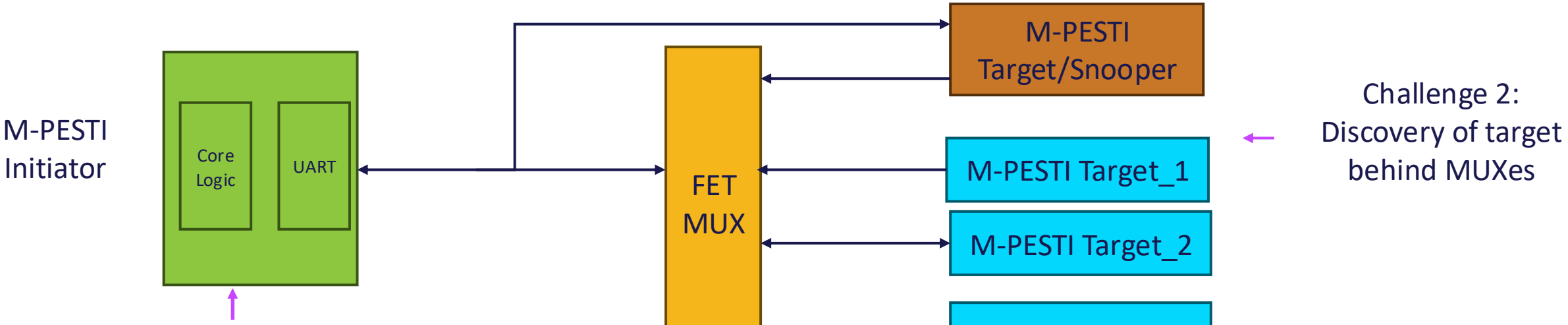
Outline

- The Modular Peripheral Sideband Tunneling Interface (M-PESTI) and the main challenges in design of the Initiator IP
- AMD’s solution to reduce M-PESTI discovery time while keeping FPGA resource utilization to a minimum
- AMD’s solution for the discovery of complex topologies
- Conclusion

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M-PESTI Initiator and Design Challenges

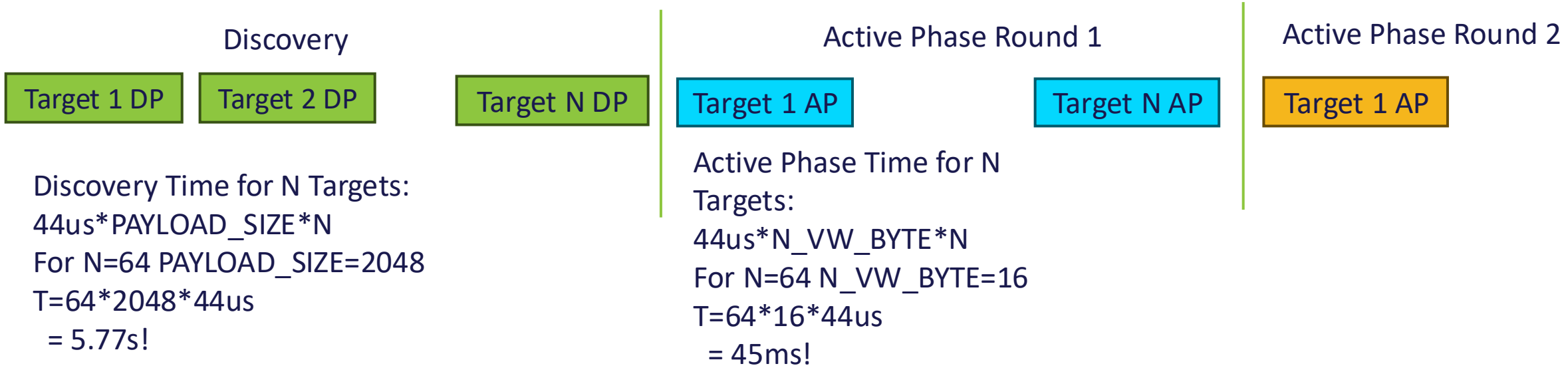
- The Modular Peripheral Sideband Tunneling Interface (M-PESTI)
  - A low-speed sideband communication protocol that enables controllers and peripherals to exchange control and status information over a half-duplex UART link



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Area Minimization: Standard's Suggestion

- The standard suggests using a single UART module in the M-PESTI Initiator design in round-robin fashion to reduce FPGA resource utilization
  - Long Discovery Time
  - Decrease BW of Virtual Wires



- More UART modules can be used: Increase in FPGA resource utilization

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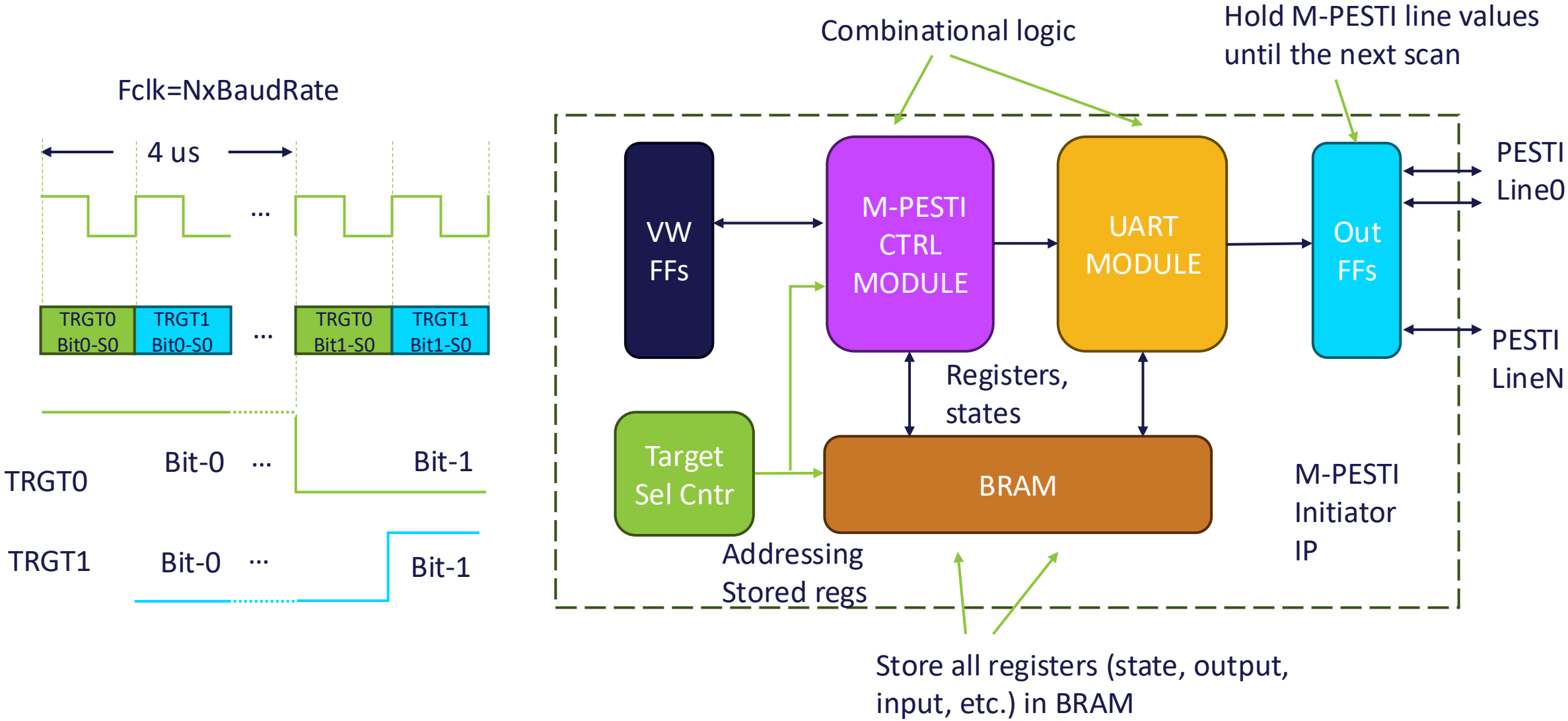
Area Minimization: AMD's Solution

- A single Control & UART IP module can be shared across **N targets** through **time division multiplexing (TDM)** when IP clock frequency is **N x the UART baud rate** with **no additional latency**
- All initiators operate simultaneously
  - The design is virtually identical to a single-initiator implementation
- Fsys=250K*N_Initiator*Sampling Rate. For 64 Initiators
  - Sampling Rate=4 => 64 MHz
  - Sampling Rate=8 => 128 MHz

} Easily Achievable in FPGA

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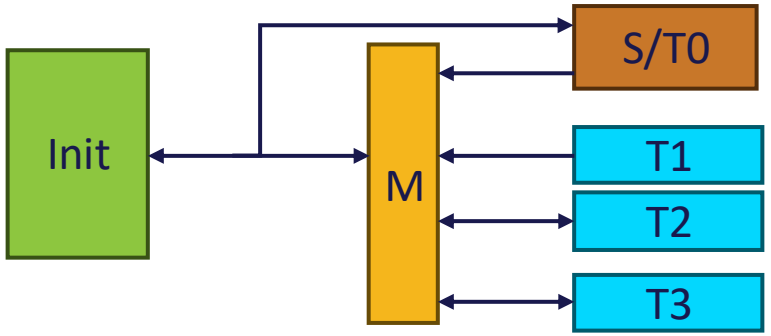
Area Minimization: FPGA Implementation



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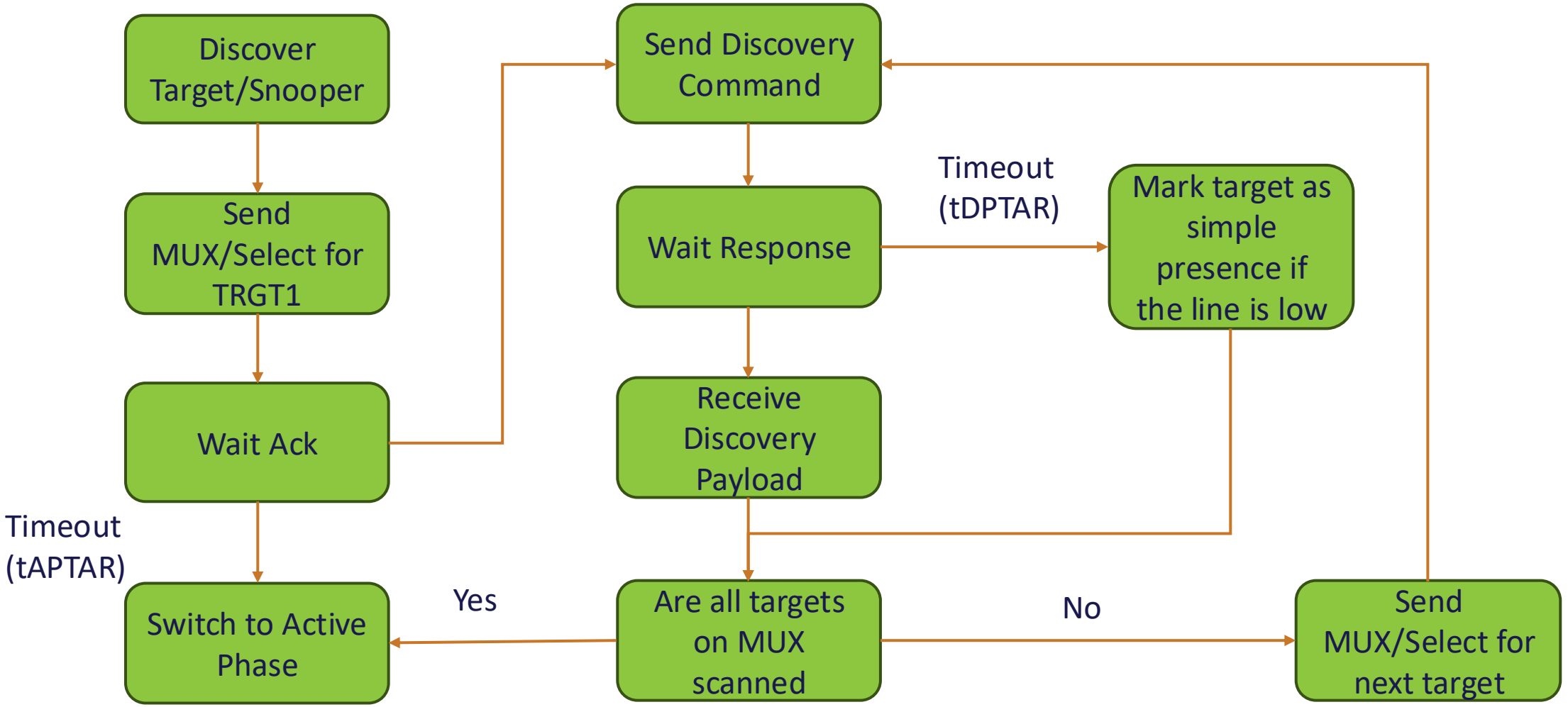
Discovery of Target Behind MUX/Switch: Break Detection

- A Break Event can occur at any moment
- The Initiator must scan targets one by one
- No guarantee that every Break Event will be detected
- Addressed in “FPGA-BEASED PESTI MUX with INTEGRATED SNOOPER, Rodriguez et al, HPE”
- Smart Target/Snooper solution suggested
  - MUX + Snooper
- Not compatible with the current standard and target designs



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Discovery of Target Behind a MUX/Switch: AMD's Approach



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Call to Action

- Provide feedback on simultaneous communication for up to 64 initiators
- Contribute to specification of target discovery behind the MUX/Switch
  - AMD’s suggestion for improvement expected in early 2027
- Specification documents on OCP Wiki:
  - https://www.opencompute.org/w/index.php?title=Server/MHS/DC-MHS-Specs-and-Designs

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